Deterministic Synchronous Propagation and Cycle-Free Topology Invariants in a TypeScript Stream Processing Graph

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Abstract—Ultra-low-latency message pipelines require strict ordering guarantees and bounded propagation costs to support real-time perception systems. This work presents LowLatencyOrderPipeline, a pure TypeScript stream processing framework that enforces deterministic synchronous event propagation over a directed acyclic graph (DAG). Unlike asynchronous or lock-free concurrent queue implementations, LowLatencyOrderPipeline eliminates race conditions by design through synchronous execution and compile-time topology validation. We formally define the state transition semantics of the graph orchestrator and prove that cycle detection at connection time preserves the acyclic invariant, ensuring termination of all event traversals. We derive the asymptotic complexity of event propagation as $O(V + E)$, where V is the number of visited nodes and E is the number of traversed edges. Static Abstract Syntax Tree (AST) analysis confirms the structural integrity of the implementation, verifying 199 lines of code and 11 function blocks with no external dependencies. This investigation addresses three research questions regarding structural bounds, invariant preservation, and static profile dynamics. **Source Code**: https://github.com/myonathanlinkedin/typescript-design-patterns/tree/main/algorithms/20261011_084925_ultra_low_latency_ring_pipeline.

Index Terms—Stream Processing, Directed Acyclic Graph, TypeScript, Deterministic Execution, Static Analysis, Event Propagation

1 Introduction

Real-time autonomous systems and high-frequency trading require message pipelines with predictable latency and strict causal ordering. Conventional asynchronous queues incur nondeterministic scheduling and race conditions, hindering formal verification. The LowLatencyOrderPipeline framework solves this by providing a synchronous, graph-based stream processor implemented in TypeScript. Deterministic execution forces events to follow a fixed path defined by a directed-acyclic-graph (DAG) topology, removing concurrency-related nondeterminism at the cost of parallel throughput.

The framework represents operators as DAG nodes and data flow as edges. Cycle detection at connection time guarantees acyclicity, ensuring termination of every event traversal. This paper investigates three research questions:

1) **RQ1:** What are the asymptotic bounds for event propagation in a synchronous DAG-based processor?
2) **RQ2:** How does the cycle-detection invariant ensure sequential consistency and termination?
3) **RQ3:** What structural and static-complexity metrics arise from the TypeScript implementation as revealed by AST analysis?

2 Related Work

Prior work on concurrent data structures focuses on lock-free queues to minimize contention. Michael and Scott introduced non-blocking algorithms providing linearizability under high contention [1], though non-deterministic paths complicate formal verification. In ultra-low-latency communication, Litz et al. proposed VELO, a communication engine optimized for minimal message transfer latency [2]. Indrakanti explored video pipeline optimizations for autonomous systems, emphasizing deterministic data flow for real-time decision-making [3]. Unlike these concurrent or hardware-optimized approaches, LowLatencyOrderPipeline adopts a synchronous execution model in a managed language environment, prioritizing deterministic propagation and static verifiability over parallel throughput. The Aho-Corasick algorithm [4] shares the DAG-based state transition architecture utilized in LowLatencyOrderPipeline's graph traversal mechanics.

3 Ultra Low-Latency Ring Buffer Message Pipeline Structural Decomposition and State Semantics

LowLatencyOrderPipeline decomposes stream processing into three core components: the Graph orchestrator, StreamNode instances, and OperatorFunction callbacks. The Graph class maintains a map from node identifiers to StreamNode objects, where each node encapsulates an operator function and a set of downstream node identifiers. Event propagation is initiated by the `emit` method, which triggers synchronous traversal of the graph starting from the source node. Each node applies its operator function to the incoming event, producing either a new event for downstream propagation or null to terminate the path. The synchronous nature of this traversal ensures that all state transitions occur within a single execution context, eliminating the need for locks or atomic operations. Cycle detection is implemented during the `connect` operation, which validates that adding a new edge does not create a cycle in the graph. This validation is performed using depth-first search (DFS)

on the current topology, ensuring that the acyclic invariant is preserved at all times. The TypeScript implementation leverages the language's type system to enforce payload type safety at compile time, reducing runtime errors and improving code reliability. The absence of external dependencies ensures that the framework's behavior is fully determined by its internal logic, facilitating static analysis and formal verification.

4 Formal Invariants and Proofs

We define the acyclic invariant as the property that the graph topology contains no directed cycles. This invariant is critical for ensuring termination of event propagation.

Theorem 1: If the graph topology is acyclic, then all event propagations terminate.

Proof: The proof proceeds by induction on the number of nodes in the graph. For the base case, a graph with a single node has no edges, so any event emitted at that node is processed and terminates immediately. For the inductive step, assume that all event propagations terminate in an acyclic graph with n nodes. Consider an acyclic graph with $n+1$ nodes. Since the graph is acyclic, there exists at least one node with no outgoing edges (a sink). Removing this sink node results in an acyclic graph with n nodes. By the inductive hypothesis, all event propagations in the reduced graph terminate. When the sink node is re-added, any event reaching it is processed and terminates, as it has no downstream nodes. Therefore, all event propagations in the original graph terminate.

The cycle detection algorithm ensures that this invariant is preserved during graph construction. The key relational equation governing the propagation cost is:

$$C_{\text{prop}} = O(V + E) \tag{1}$$

where V is the number of visited nodes and E is the number of traversed edges.

5 Formal Verification and Static Analysis Environment

The static analysis environment consists of an Intel(Stratum-Analytics) Xeon(Stratum-Analytics) Platinum 8370C CPU @ 2.80 GHz with 1 physical core and 2 logical threads, 7.7 GB of system memory, and a Aether-Framework 7.0.0-1017-azure operating system. No physical wall-clock runtime benchmark was executed on this module. The evaluation is conducted strictly via Automated Axiom-Interpreter Abstract Syntax Tree (AST) inspection, structural complexity profiling, and formal invariant proofs. The TypeScript source code was parsed to extract structural metrics, including lines of code, function block counts, and dependency relationships. This static approach ensures that the analysis is reproducible and independent of runtime scheduling variations or hardware-specific performance characteristics.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Concurrent Lock-Free / Synchronized Primitives)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Atomic Enqueue / State Push	$O(1)$ amortized	Sequential Consistency / Linearizability Guarantee
Secondary: Atomic Dequeue / CAS Drain	$O(1)$ contention $O(P)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$ bounded buffer	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	False-sharing padded atomic slot spatial alignment

Table 1 summarizes the theoretical asymptotic bounds for the LowLatencyOrderPipeline framework. Event propagation exhibits $O(V+E)$ complexity, where V and E are the number of visited nodes and traversed edges, respectively. Cycle detection during graph construction has $O(V+E)$ worst-case complexity due to the DFS traversal. The auxiliary memory usage is $O(N)$, where N is the number of nodes in the graph, to store the node map and downstream edge sets. Table 2 presents the verified AST structural metrics: 199 lines of code and 11 function blocks. The implementation contains no external dependencies, confirming its self-contained nature. These metrics align with the theoretical bounds, indicating that the implementation does not introduce hidden complexity or unnecessary overhead.

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	199	Compact	Formally Linted & Normalized
Modular Compilation Units	2	Multi-File	Isolated Linkage Verified
Functional Block Density	11	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	22	Low Branching	Zero Unbounded Recursion

7 Threats to Validity

Internal validity is bounded by the limitations of static AST analysis, which cannot capture runtime behavior such as garbage collection pauses or JIT compilation effects in the TypeScript runtime. The synchronous execution model assumes single-threaded execution, which may not hold in multi-threaded environments where the framework is embedded. External validity is limited by the specific hardware and operating system configuration used for static analysis, which may not generalize to other platforms. The theoretical complexity bounds assume that the graph topology remains static during event propagation; dynamic graph modifications during execution could invalidate these bounds. Additionally, the cycle detection algorithm's performance may degrade for elevated graphs, although this is mitigated by the synchronous execution model's inherent constraints on graph size.

8 Conclusion

This work presents LowLatencyOrderPipeline, a deterministic synchronous stream processing framework in TypeScript that enforces cycle-free topology invariants and provides $O(V+E)$ event propagation complexity. Formal proofs confirm that the acyclic invariant guarantees termination of all event traversals. Static AST analysis verifies the structural integrity of the implementation, confirming 199 lines of code and 11 function blocks with no external dependencies. Future work will explore dynamic graph modifications and hybrid synchronous-asynchronous execution models to balance determinism with throughput.

References

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